

LOAD BALANCER

High Availability | Scalability | Performance



DISTRIBUTES TRAFFIC
EVENLY



PREVENTS SERVER
OVERLOAD



AUTOMATIC FAILOVER
& RECOVERY



SSL/TLS
TERMINATION



FLEXIBLE
CONFIGURATION

★INTRODUCTION

A **Load Balancer** is a service that distributes incoming network traffic across multiple servers (such as EC2 instances) to ensure high availability, reliability, and optimal performance. It prevents any single server from becoming overloaded, improves application response time, and automatically routes traffic only to healthy servers. In cloud environments like **Amazon Web Services (AWS)**, load balancers help build scalable, fault-tolerant, and highly available applications.

□PREREQUISITES

Before you begin, ensure you have:

- An AWS account

- Basics Knowledge of AWS EC2 Security Groups, Target Groups and Load Balancing

- Internet Access to log into AWS Management Console

1. Security Group

"First, I created a Security Group. A Security Group acts as a virtual firewall that controls inbound and outbound traffic. I allowed HTTP (80) and HTTPS (443) traffic for users, and SSH (22) for secure administration."

Role: Protects the Load Balancer and EC2 instances by allowing only required traffic.

2. EC2 Instance

"Next, I launched EC2 instances. An EC2 instance is a virtual server in AWS where I deployed my web application. These instances process user requests."

Role: Hosts and runs the application.

3. Target Group

"After that, I created a Target Group and registered my EC2 instances. The Target Group performs health checks and ensures that the Load Balancer sends traffic only to healthy EC2 instances."

Role: Groups EC2 instances and monitors their health.

4. Load Balancer (ALB)

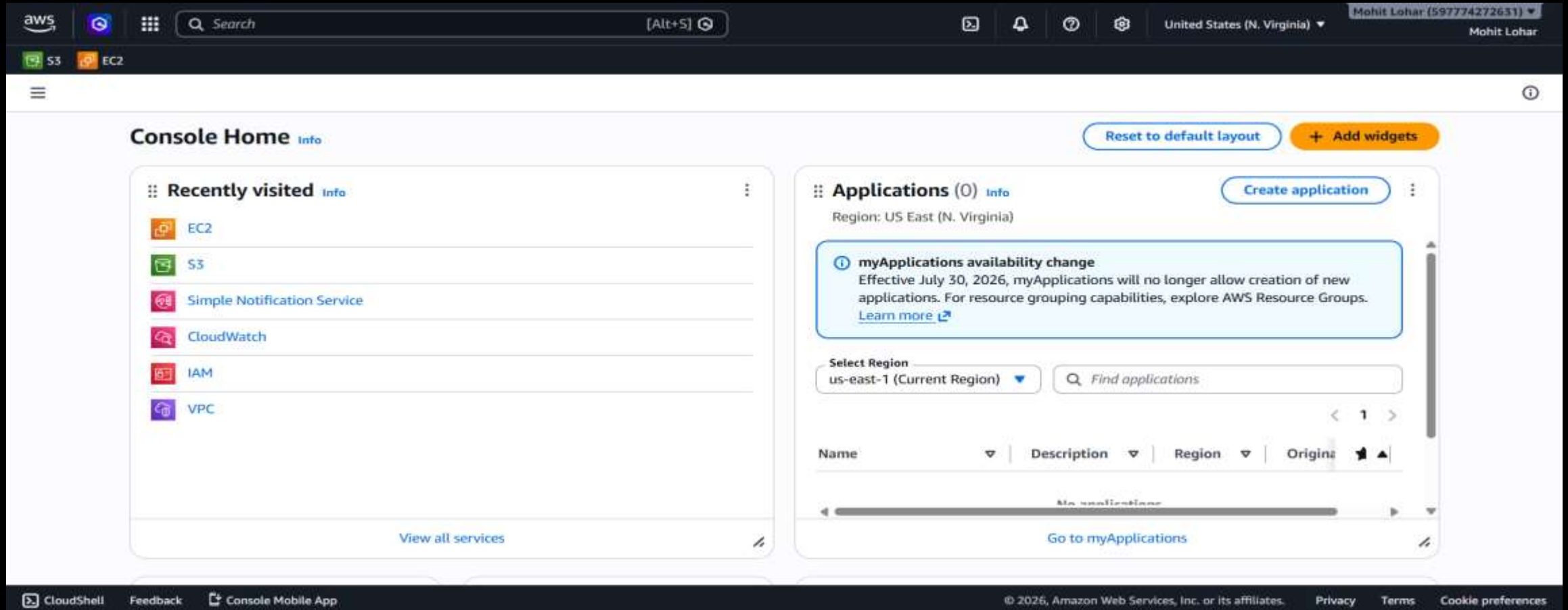
"Finally, I created an Application Load Balancer (ALB). I attached the Security Group, selected the VPC and subnets, and connected it to the Target Group. The Load Balancer receives incoming user requests and distributes them across the healthy EC2 instances."

Role: Distributes incoming traffic to healthy EC2 instances, ensuring high availability, scalability, and fault tolerance.

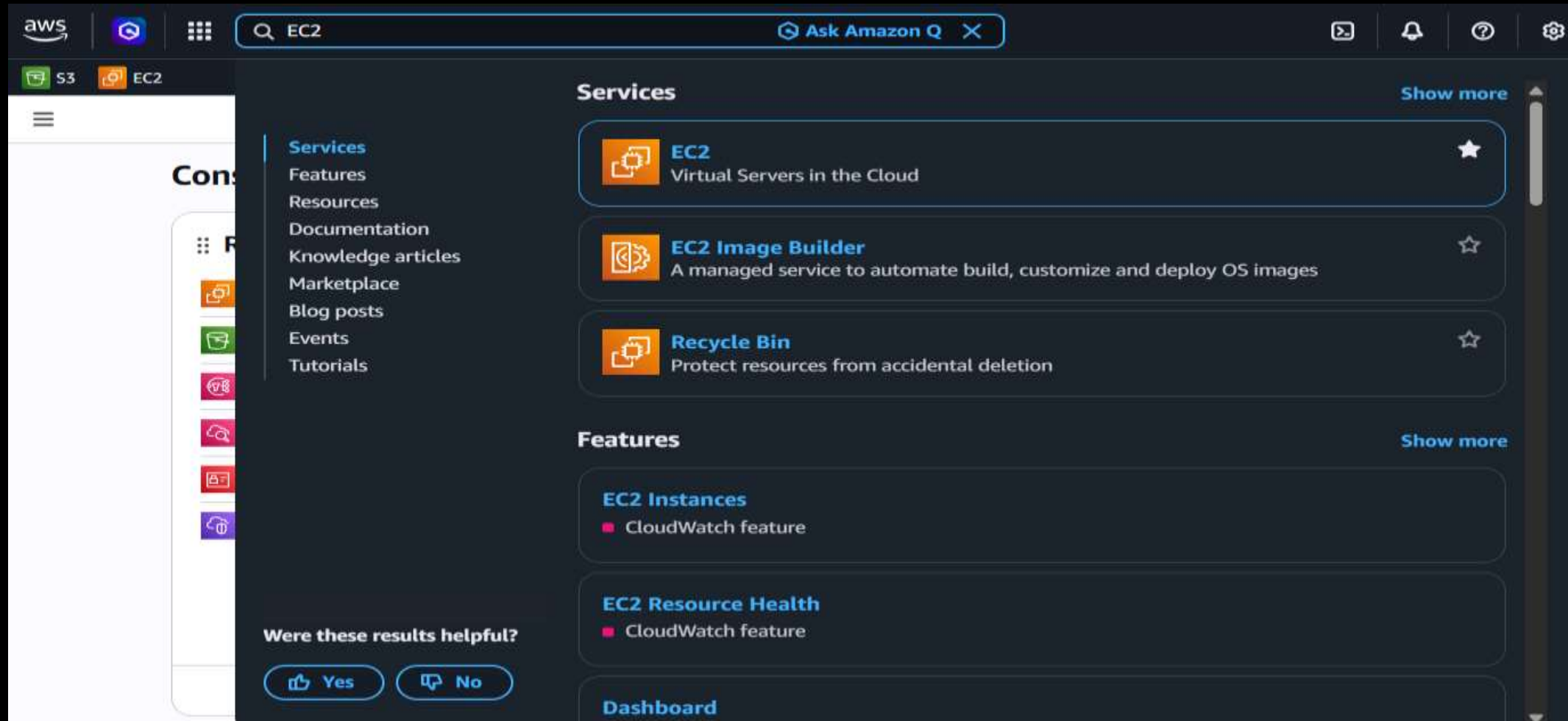
Steps to Set Up the Infrastructure

CONFIGURE SECURITY GROUP

1. Go to AWS Management Console.



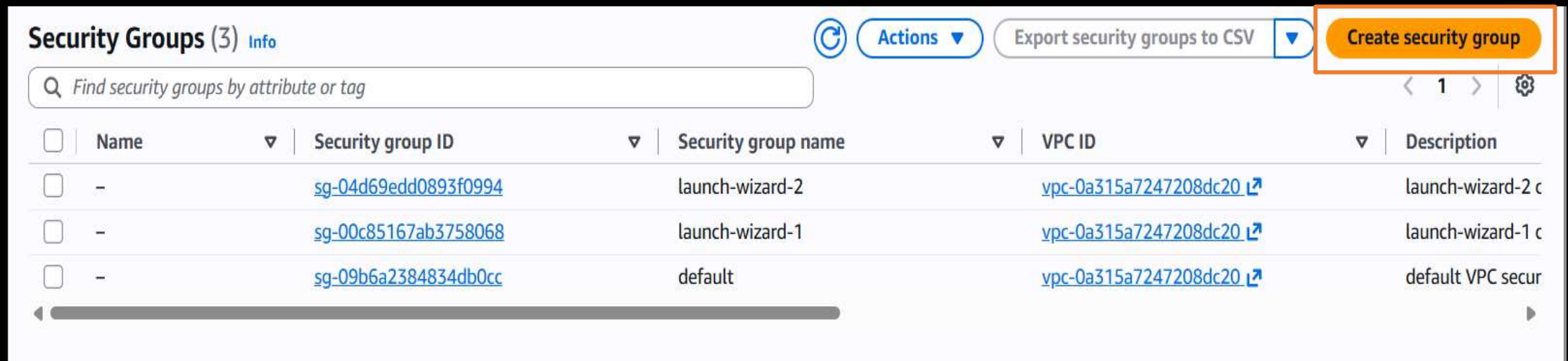
2. Search And Click EC2



3. Click on Security Group



4. Click on Create Security Group.



5. Enter Basic Details like Name and description.

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

my-SG

Name cannot be edited after creation.

Description [Info](#)

allow-ssh

VPC [Info](#)

vpc-0a315a7247208dc20

6. In Inbound Rules , add the following rules:

SSH (Port 22)

HTTP (Port 80)

HTTPS (Port 443)

Select Anywhere IPv4 as the source.

The screenshot displays the 'Inbound rules' configuration page in the AWS IAM console. It features a table with five columns: Type, Protocol, Port range, Source, and Description - optional. There are three rules listed: SSH, HTTP, and HTTPS. Each rule has a 'Delete' button to its right. Below the table is an 'Add rule' button.

Type	Protocol	Port range	Source	Description - optional	
SSH	TCP	22	Anywhere IPv4		Delete
HTTP	TCP	80	Anywhere IPv4		Delete
HTTPS	TCP	443	Anywhere IPv4		Delete

[Add rule](#)

7. Scroll Down and click on Create Security Group

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags

[Cancel](#) [Create security group](#)

✔ Security group (sg-0545773639dbb7968 | my-SG) was created successfully

▶ Details

sg-0545773639dbb7968 - my-SG

Actions ▼

Launch EC2 Instances

8. Go to the EC2 Instances Dashboard

Instances Info

Saved filter sets

Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

Last updated 5 minutes ago 

Connect

Instance state ▼

Actions ▼

Launch instances ▼

< 1 >

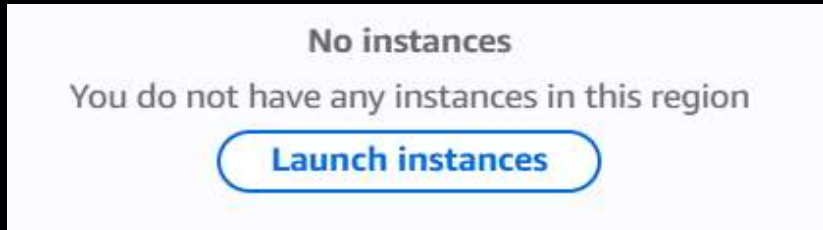


☐	Name 	▼	Instance ID	Instance state ▼	Instance type ▼	Status check	Availability Zone ▼	Public IPv4 DNS ▼
No instances You do not have any instances in this region Launch instances								

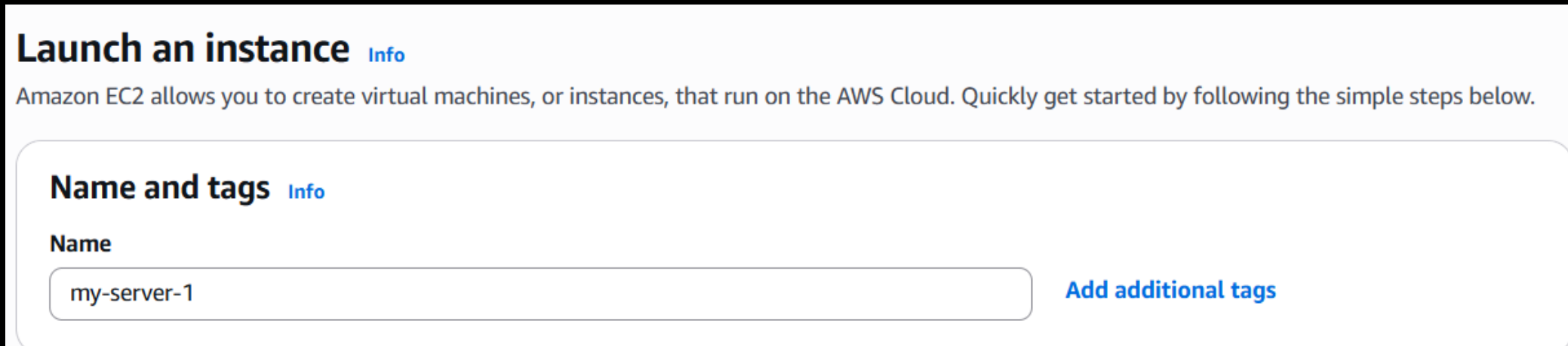
Select an instance

 ▼

9. Click on Launch Instance



10. Provide a name, Select **AMI** as **Amazon Linux**, and choose **t2.micro** or **t3.micro** (whichever is free-tier eligible).

A screenshot of the 'Launch an instance' wizard in the AWS console. The title is 'Launch an instance' with an 'Info' link. Below the title is a descriptive sentence: 'Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.' The first step is 'Name and tags' with an 'Info' link. Under this step, there is a 'Name' label and a text input field containing 'my-server-1'. To the right of the input field is a blue link that says 'Add additional tags'.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

 Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon
Linux



macOS



Ubuntu



Windows



Red Hat



SUSE Linux



Debian



[Browse more AMIs](#)

Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI

ami-06067086cf86c58e6 (64-bit (x86), uefi-preferred) / ami-071bc399fae13263b (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible



▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

11. Proceed Without a key pair:

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Proceed without a key pair (Not recommended)

Default value ▼

[↻ Create new key pair](#)

12. In Networking Setting , Select the Create security group create a earlier.

▼ Network settings [Info](#)

Edit

Network [Info](#)

vpc-0a315a7247208dc20

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-2' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere
0.0.0.0/0

☒ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

13. Scroll down to **Advanced Details** and add user **Data**.

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

 Choose file

```
#!/bin/bash
sudo yum update -y
sudo yum install -y httpd
sudo yum install -y git
export TOKEN=`curl -X PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-
metadata-token-ttl-seconds: 21600"`
export META_INST_ID=`curl http://169.254.169.254/latest/meta-data/instance-id -H
"X-aws-ec2-metadata-token: $TOKEN"`
export META_INST_TYPE=`curl http://169.254.169.254/latest/meta-data/instance-type
-H "X-aws-ec2-metadata-token: $TOKEN"`
export META_INST_AZ=`curl http://169.254.169.254/latest/meta-
data/placement/availability-zone -H "X-aws-ec2-metadata-token: $TOKEN"`
cd /var/www/html
echo "<!DOCTYPE html>" >> index.html
echo "<html lang='en'>" >> index.html
```

☐ User data has already been base64 encoded

14. Change the number of instances from 1 to 3 in the **summary** section.

▼ Summary

Number of instances

Info

When launching more than 1 instance, [consider EC2 Auto Scaling](#)

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12....[read more](#)

ami-06067086cf86c58e6

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

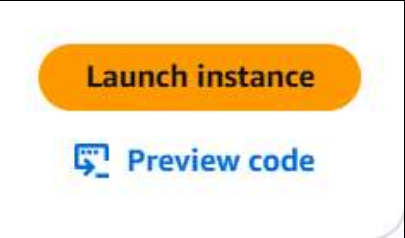
1 volume(s) - 8 GiB

Cancel

Launch instance

 [Preview code](#)

15. Click Launch Instance



16. Navigate to the EC2 Instances Dashboard, wait for the 3/3 instance check to pass.

Instances (3) Info

Last updated less than a minute ago

Connect

Instance state ▼

Actions ▼

Launch instances ▼

Saved filter sets

Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

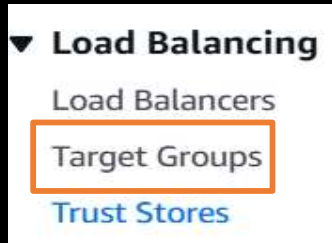
< 1 >

⚙

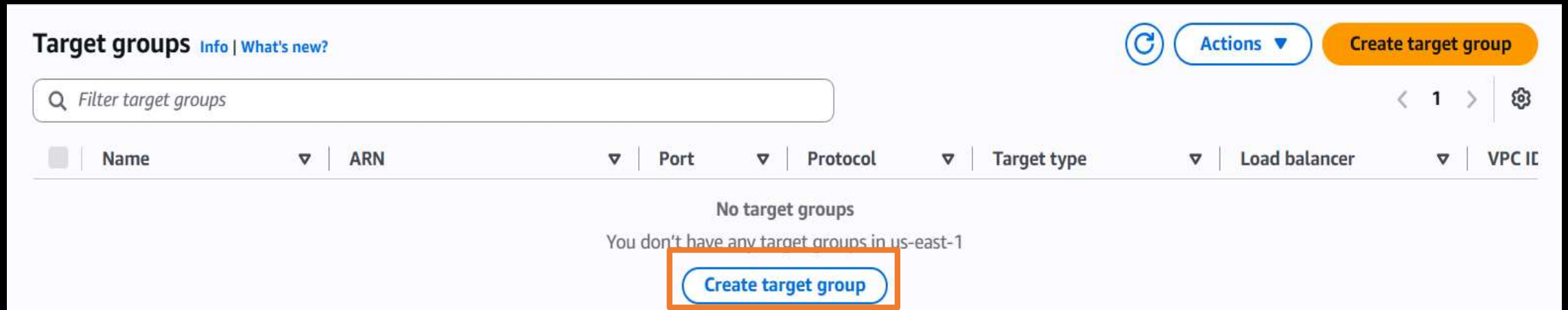
☐	Name ✎ ▼	Instance ID	Instance state ▼	Instance type ▼	Status check	Availability Zone ▼	Public IPv4 DNS ▼
☐	Select all instances	i-000bd2feff38e8a9e	✔ Running 🔍 🔍	t3.micro	🕒 Initializing	us-east-1d	ec2-13-222-201-205.co...
☐	my-server-1	i-0508e586d19e4de08	✔ Running 🔍 🔍	t3.micro	🕒 Initializing	us-east-1d	ec2-18-234-159-12.co...
☐	my-server-1	i-0a1899403b15dedf3	✔ Running 🔍 🔍	t3.micro	🕒 Initializing	us-east-1d	ec2-204-236-200-239.c...

Create the **TARGET GROUPS**

17. In the **Load Balancing** Section, click on **Target Groups**.



18. Click **Create Target Group**.



19. Enter a Target Group name, then click Next.

Target group name
Name must be unique per Region per AWS account.

my-TG

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 5/32

Protocol
Protocol for communication between the load balancer and targets.

HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

Attributes

 Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

 **Tags - optional**

Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

Cancel

Next

ELB PROJECT DOCUMENTATION

BY MOHIT GADILOHAR

20. Select available instances and Register them.

Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (3/3)

Filter instances

☒	Instance ID	Name	State	Security groups	Zone
☒	i-0beaa14bf23c8cf22	my-server-1	Running	launch-wizard-1	us-east-1d
☒	i-01b5b5f8ad88bd6fb	my-server-1	Running	launch-wizard-1	us-east-1d
☒	i-06e22706e1b1e8771	my-server-1	Running	launch-wizard-1	us-east-1d

3 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

21. Review Target and click Create Target Group.

Step 2: Register targets

Edit

Targets (3)

Instance ID	Name	Port	Zone
i-0beaa14bf23c8cf22	my-server-1	80	us-east-1d
i-01b5b5f8ad88bd6fb	my-server-1	80	us-east-1d
i-06e22706e1b1e8771	my-server-1	80	us-east-1d

► Server-side tasks and status

After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

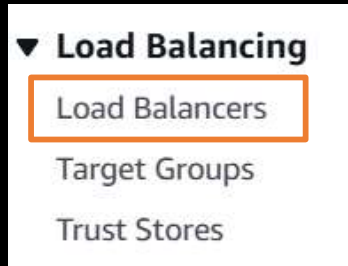
Cancel

Previous

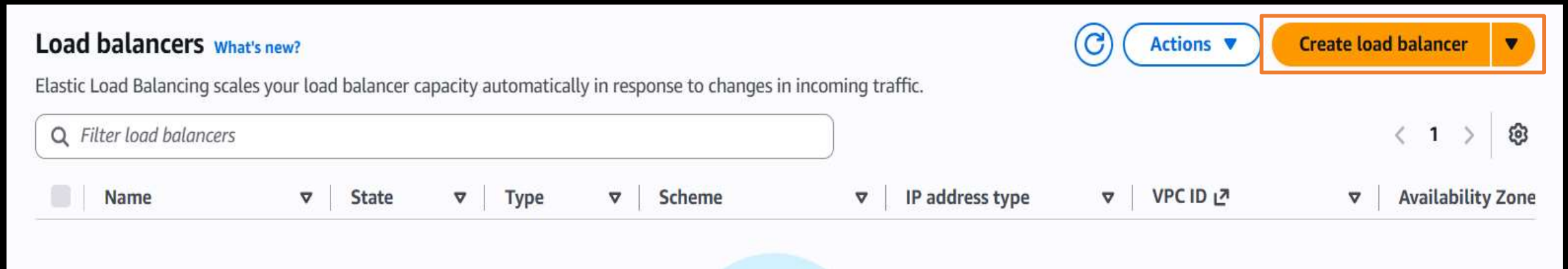
Create target group

SET UP ELASTIC LOAD BALANCER (ELB) ▮

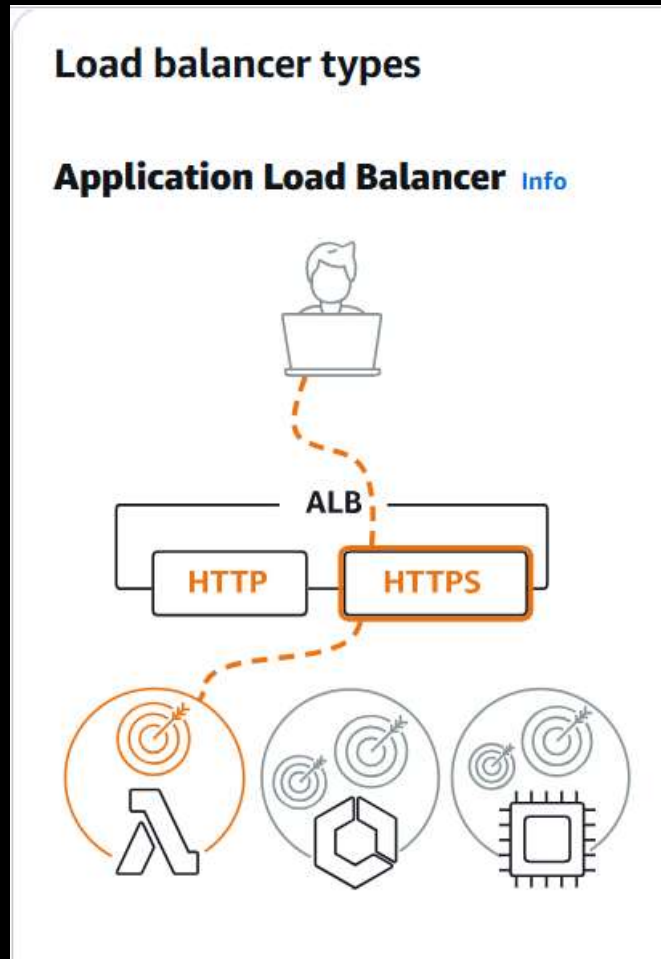
22. Go to Load Balancers in the Load Balancing sections



23. Click Create Load Balancer.



24. Choose the Application Load Balancer (ALB) and click Create



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

[Create](#)

25. Enter the name for the Load Balancer.

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

my-LB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme | [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves Internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

26. In Network Mapping choose the Availability zone.

Network mapping | [Info](#)
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC | [Info](#)
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0a315a7247208dc20
172.31.0.0/16

(default)  [Create VPC](#)

IP pools | [Info](#)
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view **Pools** in the [Amazon VPC IP Address Manager console](#).

☐ **Use IPAM pool for public IPv4 addresses**
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets | [Info](#)
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☐ **us-east-1a (use1-az1)**

☐ **us-east-1b (use1-az2)**

☐ **us-east-1c (use1-az4)**

Availability zone

☒ **us-east-1d (use1-az6)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-0668063827a03cc81
172.31.32.0/20
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

☒ **us-east-1e (use1-az3)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-08f79ac97e22e6580
172.31.48.0/20
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

☒ **us-east-1f (use1-az5)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-0259fb7bd4a358bf6
172.31.64.0/20
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

27. In Security Group remove the default Security Group and attach the Security Group created earlier

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups

Select up to 5 security groups

default (sg-09b6a2384834db0cc) X

Inbound rule (1) Outbound rule (1)

Create security group [↗](#)

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups

Select up to 5 security groups

Create security group [↗](#)

Application Load Balancers require at least one security group. If none are selected, the VPC's default security group will be applied.

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Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups

Select up to 5 security groups

Q |

☐ my-SG (sg-093a634a43e7c85f9)
Inbound rule (3) Outbound rule (1)

☐ launch-wizard-2 (sg-0adfc82596796672f)
Inbound rule (3) Outbound rule (1)

☐ default (sg-09b6a2384834db0cc)
Inbound rule (1) Outbound rule (1)

☐ launch-wizard-1 (sg-0ed6206c4f720786c)
Inbound rule (3) Outbound rule (1)

my-SG (sg-093a634a43e7c85f9)

Create security group

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups

Select up to 5 security groups

my-SG (sg-093a634a43e7c85f9) X
Inbound rule (3) Outbound rule (1)

Create security group

28. In Listeners and Routing, attach the Target Group created earlier.

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80

Remove

Protocol

HTTP ▼

Port

80

1-65535

Default action [Info](#)

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups

☐ Redirect to URL

☐ Return fixed response

Forward to target group [Info](#)

Choose a target group and specify routing weight or [create target group](#).

Target group

Select a target group ▼

Ⓢ

Weight

1

0-999

Percent

100%

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Target group

Select a target group ▲

Q

my-TG
Target type: Instance, IPv4 | Target stickiness: Off

HTTP

⌂

Weight

1

0-999

Percent

100%

Target group stickiness

Info

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Target group

my-TG
Target type: Instance, IPv4 | Target stickiness: Off

HTTP ▼

⌂

Weight

1

0-999

Percent

100%

29. Scroll down and click **Create Load Balancer**.

Service integrations [Edit](#)

Amazon CloudFront + AWS Web Application Firewall (WAF): -
AWS WAF: -
AWS Global Accelerator: -

Tags [Edit](#)

-

Attributes

Certain default attributes will be applied to your load balancer. You can view and edit them after creating the load balancer.

Creation workflow and status

► **Server-side tasks and status**

After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

[Cancel](#)

Create load balancer

30. Wait for the ELB status to become Active

my-LB

Actions

▼ Details

Load balancer type

Application

Scheme

Internet-facing

Status

Provisioning

Hosted zone

Z35SXDOTRQ7X7K

VPC

[vpc-0a315a7247208dc20](#)

Availability Zones

[subnet-08f79ac97e22e6580](#)

 us-east-1e (use1-az3)

[subnet-0668063827a03cc81](#)

 us-east-1d (use1-az6)

[subnet-0259fb7bd4a358bf6](#)

 us-east-1f (use1-az5)

Load balancer IP address type

IPv4

Date created

July 5, 2026, 13:59 (UTC+05:30)

Load balancer ARN

arn:aws:elasticloadbalancing:us-east-1:597774272631:loadbalancer/app/my-LB/6c62832448aaad69

DNS name

Info

my-LB-1969767601.us-east-1.elb.amazonaws.com (A Record)

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

ELB PROJECT DOCUMENTATION

BY MOHIT GADILOHAR

my-LB



Actions ▼

▼ Details

Load balancer type

Application

Scheme

Internet-facing

Status

✓ Active

Hosted zone

Z35SXDOTRQ7X7K

VPC

[vpc-0a315a7247208dc20](#) ↗

Availability Zones

[subnet-08f79ac97e22e6580](#) ↗ us-east-1e (use1-az3)

[subnet-0668063827a03cc81](#) ↗ us-east-1d (use1-az6)

[subnet-0259fb7bd4a358bf6](#) ↗ us-east-1f (use1-az5)

Load balancer IP address type

IPv4

Date created

July 5, 2026, 13:59 (UTC+05:30)

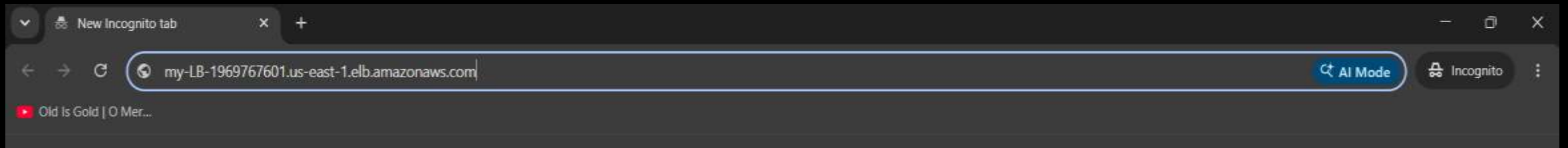
Load balancer ARN

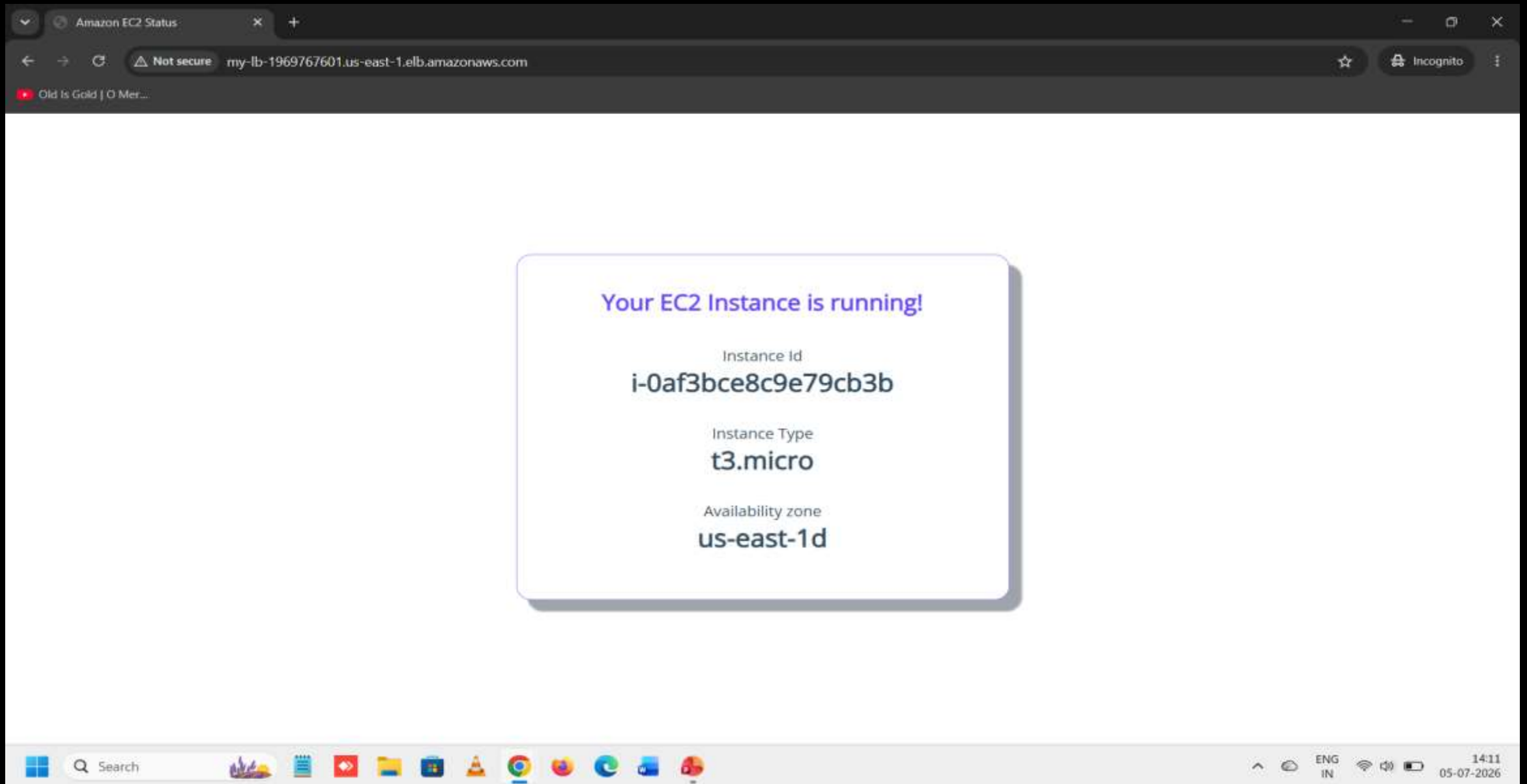
 `arn:aws:elasticloadbalancing:us-east-1:597774272631:loadbalancer/app/my-LB/6c62832448aaad69`

DNS name [Info](#)

 `my-LB-1969767601.us-east-1.elb.amazonaws.com` (A Record)

31. Copy the **DNS mode** And Go to the Incognito window and paste **DNS mode**.





32. Click **Refresh Button**. And see the **Instance rotate**.

Your EC2 Instance is running!

Instance Id

i-0af3bce8c9e79cb3b

Instance Type

t3.micro

Availability zone

us-east-1d

Your EC2 Instance is running!

Instance Id

i-080bc45f19407df4b

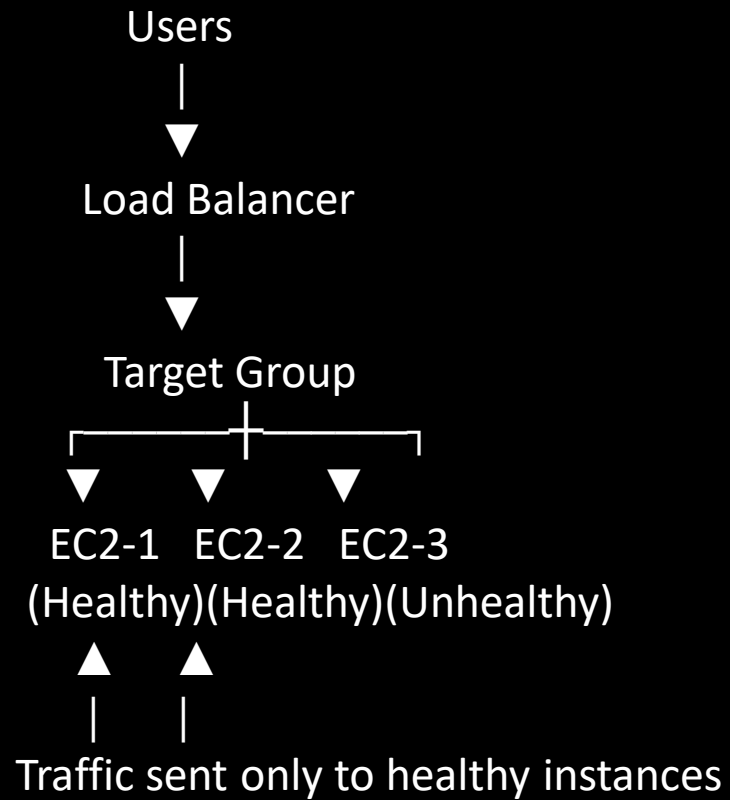
Instance Type

t3.micro

Availability zone

us-east-1d

Architecture



Security Group protects:

- ✓ Load Balancer
- ✓ EC2 Instances



Thank You!

— ♥ YOUR SUPPORT MEANS A LOT ♥ —